One Sentence at a Time: A Quantitative History of Rationality in Economic Thought

ARTICLE HISTORY
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Abstract
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rationality; economics; bibliometrics; text mining; natural language processing; large language models

JEL CLASSIFICATION
B00; B20

Warning

This is a work in progress. The discussion Section in particular will be expanded in future versions.

Interactive application to explore the data

We have developed an interactive application to explore the data and results presented in this paper. It is available at: https://0199205a-7b17-ab11-595d-8f12aae160dd.share.connect.posit.cloud. The first tab allows to explore the bibliometric networks; the second tab is a global summary of textual clusters.

Introduction

In the last decade, the number of history of economics papers employing quantitative methods has increased (see e.g., Goutsmedt, Claveau, and Herfeld 2023 special issue). Several essays have adopted a reflexive stance, examining the implications of using quantitative methods in the history of economics (Cherrier and Svorenčík 2018; Edwards, Giraud, and Schinckus 2018). These contributions have aimed to provide broad discussions on the use of such methods. However, given the virtually unlimited variety of quantitative approaches potentially relevant to historians of economics—and the wide array of research questions they can address—these reflections often remain abstract and offer little practical guidance. As a result, they tend to provide useful broad overviews, but without clearly articulating what quantification entails in practice, and how it can be both meaningful and challenging to integrate into historical research.

Our contribution sits between a historical study that uses quantitative methods to answer a specific question and a general methodological discussion of those methods. From the collection of data to the interpretation of results, we illustrate concretely how these methods are useful and examine, in practice, the methodological choices they entail. We present a step-by-step application of specific quantitative methods to a broad topic: the history of rationality in the 20th century.

We focus our discussion on unsupervised methods that have been particularly influential in the history of economics. Unsupervised methods are machine-learning techniques in which algorithms identify patterns from unlabelled data, as opposed to supervised learning methods—such as regressions—that learn patterns from labelled data to make inferences. Their goal is not necessarily to provide a "measure" (Grimmer, Roberts, and Stewart 2022)—though they can be adapted to do so—but to organise in categories large corpora and enable accelerated and "distant reading" (Moretti 2013; Guldi 2023). They are well suited to historical inquiry because their unsupervised nature reduces the risk of presentism: rather than imposing present-day categories on the past, unsupervised algorithms uncover patterns directly from the data. When time is explicitly incorporated, they allow researchers to map the discipline at different points of time, to identify the emergence and decline of subjects and concepts, and to assess the influence of specific economists.

Our article uses two types of data, texts and citations, that have been particularly effective in recent quantitative studies, texts and citations. The analysis of textual corpora offers a direct window into the semantic content of debates, while citation data, on the other hand, provides a lens through which to view intellectual interconnections and structures of influence within the discipline. We show how a large corpus of documents can be classified based on the information contained in textual and citation data—what we call "textual clusters" and "bibliometric communities."[1]

Above all, our discussion aims to illustrate a core principle for historical inquiry with such unsupervised quantitative methods. They require continuous back-and-forth between aggregate quantitative results, complementary indicators used for interpretation, and preliminary knowledge and close reading of primary sources. These methods function not only as a form of corroboration but also as "discovery methods" (Grimmer, Roberts, and Stewart 2022): they facilitate the exploration of large datasets to reveal historical patterns. They often confirm, and sometimes complement, established findings. They can also reveal pitfalls and blind spots in existing research.

The idea of "rationality" is an effective focus for a concrete demonstration. First, many historians of economics engage with it in one way or another, which makes the exercise relevant for a broad audience. Second, because the concept is broad and pervasive in economics, applying quantitative methods to a very large corpus is particularly informative.[2] We use a corpus of around X articles extending back to 1900 to show how these methods can handle long time horizons. Third, the multiple meanings attached to "rationality" and its uses applied to various subjects show how textual methods, coupled with bibliometrics, can help capture this semantic plurality.

In what follows, we concentrate on what such a quantitative analysis requires in practice. We

[1] The bibliometric communities, identified through network analysis, could also be called clusters. But we opted for "communities" in order to distinguish them from the "textual clusters".

[2] To be clear, we don't think that quantitative methods are only helpful for very large corpora. However, it is where their surplus value may appear as the more obvious.

highlight what kinds of questions and challenges arise at each stage of the research process. This focus lets us highlight (a) the crucial issue of selecting sources and building data; (b) how even simple quantitative assessments can be informative; (c) the challenges and subjective choices involved in building and adapting tools; and (d) why interpreting results demands careful attention to various indicators and close knowledge of both the corpus and its historical context. By tracing, step by step, how we assemble and analyze our corpus, we aim to provide a practical example and a reflection on broader methodological issues faced by quantitative historians of economics.

From sources to corpus

From sources to data

Discussions on quantitative methods often focus on the varieties of existing methods. In practice, however, analysis and interpretation come last. Most effort goes into collecting, cleaning, and structuring data, much of which remains invisible in published work. Sources rarely arrive as ready-to-use datasets; they must be transformed from somewhat raw materials into usable corpora. In short, the quantitative historian must be as much a data wrangler as a data analyst.

Bibliometric databases are a convenient way to access corpora of economic texts: they record relatively well-structured data, gathering key features of scientific output—such as authors, journals and affiliations, *etc.*. Some of these databases, such as *Web of Science, OpenAlex,* or *Scopus,* record citation data, which enables tracing intellectual influence through diffusion patterns, and mapping scholarly contributions over time.

Each database has its own strengths and weaknesses. While at a very general level Web of Science, OpenAlex, or Scopus have similar coverage (**martinGoogle2021?**; Culbert et al. 2025), some studies might suffer from choosing a less appropriate database regarding the research questions. For instance, Scopus has a lower coverage of the top 5 economics journals before the 1990s and while being openaccess, OpenAlex has been less curated than the products of private publishers. Whatever the provider, less successful or now-defunct journals are also more likely to have incomplete or discontinuous digital coverage, impeding their inclusion for historical analyses. Additionally, databases are oriented toward English-speaking contributions; these databases may thus be deficient for a research project targeting another or multiple languages.[3] Last but not least, although they provide useful metadata and citation information, these databases generally lack full text, which is subject to copyright and therefore cannot be obtained from a single publisher.

Even when citations and full texts are available, the amount and quality of information that can be reliably encoded remain limited. Citation data are notoriously difficult to structure because both the notion of what constitutes a reference and the conventions for recording it have changed over time. As a result, an article may extensively cite a document—such as a working paper or a report—that does not follow standard bibliographic formats and thus never appears in citation databases. Full texts face similar limitations: mathematical expressions and empirical material, such as tables, are often poorly captured or inconsistently encoded by providers. These shortcomings restrict what can be studied by a quantitative analysis.

A first important point is, therefore, that data availability shapes what we are able to ask and so to answer. The scarcity and structure of available data affect every

[3] One of the main challenges for a quantitative history of economics is to move beyond reliance on proprietary digital libraries and to actively develop open, historically inclusive corpora. This includes incorporating materials produced in non-Anglophone countries and expanding the range of textual formats considered—such as books, book reviews, working papers, and conference proceedings.

stage of inquiry, from the choice of research questions to the interpretations of quantitative results. In practice, research may be shaped as much by the scarcity of our data as by researchers' own preferences. It is of course possible to build handmade datasets from scratch, but, in the case of textual and citation data, such tasks remain time-consuming beyond small-scale study. More commonly, it is often necessary to combine information from existing databases to fulfill a particular goal. For example, even for a medium-scale study of the publications of the European Economic Review (Goutsmedt and Truc 2023)—few thousands documents—required to combine three heterogenous databases: Econlit (for JEL codes classification) Web of Science and Scopus (for missing coverage of Web of Science). In their study of the French journal *La Revue Economique,* (**DelceyRevue2025?**) had to enrich the dataset provided by the journal with missing full texts and authors' information from the digital libraries Persée, Cairn, and IdRef.

For our study of rationality, the first step was to identify the best sources. For citation data, the Web of Science provides the most reliable and consistent coverage for our period and has been widely used in the history of economics. As for full text, we relied on three providers. First, JSTOR's full-text collection offers high-quality scans of most leading economics journals. Second, we used Scopus to identify peer-reviewed economics journals not included in JSTOR and collect a new list of articles. We retrieved full texts first through the Elsevier Full-Text API, when available. Finally, remaining full texts were obtained through the ISTEX project, which provides access to a substantial corpus of documents for researchers affiliated with French universities.[4]

The Figure 1 shows the distribution of this corpus. It can be already noted that our corpus disproportionately represents Anglo-Saxon journals, which are also those most systematically digitized and preserved. Not only do we tend to overlook other traditions of research, but this also raises a serious issue of presentism: the fact that retrospective data on these journals are easily accessible today does not imply that they were equally central in earlier ones, nor that articles were the dominant vehicle for the diffusion of ideas. Rather, it reflects the fact that they had the resources and opportunities to digitize and maintain comprehensive digital archives of their publications.

Once access to full text is secured, another crucial step is to transform them into a usable database. While Web of Science citations are already delivered in a relatively structured form, full texts require substantial processing. In most cases, data are not given but result from a process that involves cleaning, categorizing, and selectively removing or reformatting information from raw sources in order to produce a dataset suitable for computational analysis. This challenge is particularly true in the history of economics, where the primary sources are the texts themselves—often unstructured and interspersed with various layers of metadata, such as section headings, footnotes, bibliographic references, or editorial annotations. This requires making important choices well before the actual stage of analysis.

Our first choice was to restrict the analysis to English-language articles, since cross-language comparisons involve additional challenges, which would go far beyond the scope of this article.[5] We also restrain our analysis to research articles and filter out book reviews, comments, editorial reports or obituaries.[6] While such materials can illuminate how rationality was debated, they are not primary sites for articulating new ideas within the discipline. Moreover, textual data are by nature voluminous, and filtering improves tractability for large-scale computation. At the document level, we focused on the body text

[4] For more detailed information on the collection of data, see the appendix.

[5] Indeed, most textual methods are designed to identify commonalities within texts that share a specific linguistic structure. Hence, the same topic discussed by two documents in two different languages will likely not be identified as related because linguistic differences mask underlying semantic similarity.

[6] This filtering draws on JSTOR's and Scopus' own classifications, supplemented by some additional filtering from ourselves. Despite these safeguards, a perfectly clean restriction to research articles was not feasible, and some residual non-article items likely remain.

and removed (as far as possible) peripheral elements such as acknowledgements, references, or appendices. We also focus on natural-language text and remove other information that is poorly OCR-processed and often unusable, such as mathematical formulas and empirical tables.

This data wrangling is not a tedious prelude to "real" historical analysis. Manipulating, cleaning, and structuring raw sources is a fundamental and often generative stage of research. As Lemercier and Zalc (2019, 62) reminds us, collecting and categorizing sources is also "a moment to reflect on the sources and the purpose of the research." Direct engagement with raw data can prompt the reevaluation of initial hypotheses and the emergence of new questions. In our case, preparing full-text documents raises basic design choices about inputs and, by extension, what counts as a relevant economic text. Should we include book reviews, editorials, working papers, or conference proceedings? How should we define an "economics journal"—restrict it to core outlets or extend it to interdisciplinary venues where economics appears regularly? Even within a single article, boundaries are ambiguous: should abstracts, footnotes, appendices, or acknowledgments be analyzed or excluded? Each decision may carry historiographical implications. It shapes the corpus and, ultimately, the history of rationality our methods can reveal. There is rarely a single definitive and uncontestable choice, but rather a series of trade-offs that should be made explicit and required to engage with some of the material available.

To give a specific example, it is only after a first batch of exploratory analysis that we notice that some important economics journals were missing in JSTOR. If it was obvious from preliminary results that the emergence of behavioral economics impacted how economists discussed rationality, from our own expertise on the matter, we observed that some journals like the *Journal of Economic Behavior & Organization* or the *Journal of Behavioral Economics*—later to be the *Journal of Socio-Economics*—were missing in places where they should be predominant. This prompted us to further explore potential biases in JSTOR and to augment our corpus with new full texts extracted from Elsevier and the ISTEX project. Exploring raw sources is thus an important step that requires to already engage with your material and the existing literature.

The use of several databases also raised specific issues. Here, our goal was to combine our textual data (from the three providers mentioned above) with citation data from WoS. Achieving this required linking documents across datasets, yet WoS—like many databases—does not provide DOIs, which would otherwise serve as convenient unique identifiers. It therefore fell to us to develop matching procedures to determine whether an article retrieved from JSTOR or Scopus corresponded to the same article indexed in WoS. However, matching based solely on the title is unreliable. For example, the title *"Inflation and Unemployment"* may refer either to James Tobin's AEA presidential lecture or to Milton Friedman's Nobel lecture. Consequently, we had to supplement title information with additional metadata, including the journal name—which required standardizing journal titles across databases—as well as the publication year, volume, issue, and page range. Differences in journal coverage and in data storage practices (particularly title formatting) mean that it is generally impossible to achieve a complete match between databases. In our case, approximately one-quarter of the full-text documents could not be matched to WoS records, which prevents us, when relevant, from analyzing their citation data.[7]

From data to corpus

In parallel with the transformation of sources into data, we also needed to establish the boundaries of our corpus, a process that involves selecting materials either *ex ante*, when choosing which sources to include, or *ex post*, when filtering the collected data. In the context

[7] See the appendix for more information on the matching process.

of our project on the history of rationality in economics, this required us first to determine what qualifies as "economics," and second to identify which documents or parts of documents can be considered "texts" about rationality.

The first challenge was thus to define the extent of our corpus a priori, i.e. to delineate economics as an object of study. Some research objects are relatively easier to delineate, and the transition from a database to a well-defined corpus is therefore straightforward. For example, writing the history of a particular journal (Edwards 2020; **DelceyFama2025?**) or of one or several individuals (Truc 2025) entails comparatively definitional or boundary challenges. In contrast, most objects studied by historians lack clear definitions or boundaries. While some large-scale studies focus on the discipline as a whole (**claveauMacrodynamics2016?**; **trucInterdisciplinarity2023?**; Ambrosino et al. 2018), such work still requires an operational definition of what are documents in "economics." This definitional issue becomes even more pronounced when the object of study is a specific "field" (Cherrier, this issue) .

Quantitative contributions force the researcher to settle on a narrow but operational definition of the object under study. Non-quantitative historical methods, by contrast, tend to address large objects by focusing on their "core" rather than their boundaries: historians of economics typically reconstruct the history of materials that are unambiguously part of the object under study, thus making a narrow definition often unnecessary. Quantitative approaches, however, require a well-defined corpus and consequently oblige researchers to impose simple, clear-cut boundaries on complex and ambivalent categories such as "economics". Defining the relevant historical materials therefore requires the adoption of specific conventions to approximate the object under study (**DesrosieresPolitique_1993?**). Therefore, defining a corpus constitutes a quantification convention—one that must always be assessed instrumentally, not as an absolute definition but as an operational hypothesis serving a specific research question.

Many proxies have been used in the history and philosophy of economics to define disciplines and sub-disciplines. For instance, (Goutsmedt and Truc 2023) used the JEL codes to select macroeconomic documents in a corpus of well-established economics journals, while (Truc 2022) and (Jullien and Truc 2024) relied respectively on citations data and institutional affiliations to identify the boundaries of behavioral economics.[8] To choose a particular proxy, it is important to understand how it relates to the object under study. Both JEL codes and journal classifications—like JSTOR or Scopus classifications—can structure a corpus in ways that reflect their own histories, and researchers must therefore consider how these proxies shape the boundaries of their material. For example, while the JEL codes for neuroeconomics emerged around the same time as the first publications in the field, the JEL codes for behavioral economics appeared more than two decades after the earliest contributions (Truc 2023). Understanding the historical development of such proxies is thus essential for interpreting the nature of the corpus they produce. More generally, a thorough knowledge of the history of the object being studied is crucial for evaluating the adequacy and representativeness of the resulting corpus.

In our case, we first define economic documents by building the corpus from peer-review journals. This convention has both advantages and limitations. On the one hand, this approach is unambiguous, as it relies on agreeing upon a predefined list of journals, rather than, for instance, determining at the document level which individual articles qualify as economics. Moreover, journals constitute one of the main legitimate institutional markers of a discipline, alongside training programs and professional associations. On the other hand, focusing on journals means excluding other publication outlets, such as books or working

[8]X study fragmentation of economics through the "blue ribbon" of economics journals… (OeConomia special issue)

papers.[9] In addition, it struggles to capture interdisciplinarity, whether in the form of interdisciplinary journals or of economists publishing in other disciplinary journals. Finally, this strategy requires the use of arbitrary criteria to decide which journals to include and which to exclude—particularly for journals at the margins of "economics", either because of their interdisciplinary orientation or because of uncertainty regarding their peer-review standards or academic practices.[10] We eventually settled on a carefully curated list of 329 journals identified as economics journals in JSTOR and Scopus, but excluding journals that are not entirely academic, insufficiently focused on economics, or only founded within the last twenty years.[11] We were able to retrieve 337,086 articles published in these journals from 1900 to 2009.

The second challenge was to restrain our corpus to documents engaging with the issue of rationality. One of the most straightforward proxies are keywords (Truc 2023). Based on a predefined list of target terms (often called a "dictionary"), this approach restricts a corpus to documents that mention these terms with at least a given frequency. In addition to its relative simplicity, it works well for specific and unambiguous terms, like "stagflation" and "Great Inflation" (Goutsmedt 2021) or "Agent-Based Models" (Baccini et al. 2025). But this approach may display several limitations when it is not the case. Indeed, it focuses on spotting occurrences of a term rather than a general idea. Just think about the various ways rationality could be discussed in the history of economics: beyond the term of "rationality" itself, economists employ various expressions that refer to close ideas such as maximising profits or expected utility, the *homo oeconomicus*, or the transitivity or completeness of preferences.

Going beyond simply searching for occurrences of "rationality" and "rational," we could have constructed an extensive dictionary of terms associated with the concept of rationality. This task requires substantial knowledge of the history of economics and of debates surrounding rationality, and thus presupposes that researchers possess adequate historical and conceptual background. Despite this, it remains difficult to prevent the dictionary-building process from introducing biases—for instance, by omitting concepts that are historically relevant but salient only during specific periods (such as "hedonism"), or by creating a disproportionate dictionary, with many terms related, for instance, to decision theory but few pertaining to macroeconomics or public economics. Consequently, this approach also constrains the potential for discovery: by setting the boundaries of the dictionary in advance, researchers may unintentionally exclude terms or themes of which they were unaware, and thus remain unaware of them throughout the analysis.

To overcome this issue, we use a Large Language Model (LLM) to identify documents—and even specific sentences within documents—that deal with rationality in our corpus of economic articles.[12] LLMs are algorithms trained on extremely large collections of text to learn patterns in language. In simple terms, they learn to predict missing words in a sentence or to determine whether two sentences follow each other. Through this training, the model develops billions of internal parameters that capture regularities in vocabulary, grammar, and meaning. When we input text into such a model, it translates sentences into vectors that summarize their context and meaning. This enables us to compare sentences not by the exact

[9] Adding these other outlets would pose additional challenges. For both working papers and, above all, books, no large-scale databases provide comprehensive full texts or complete reference lists. In addition, working papers raise the problem of potential double counting, as they may eventually be published in economic journals.

[10] Pottier et al. were confronted to this point when studying… [More general references about the explosion of the number of journals and the questions it raised in studying specific field…]

[11] We did not filter journals by language a priori. Indeed, many national non-anglo-saxon journals also published in English at some more or less frequent occasions. It was thus easier to filter by language a posteriori and at the document level, once articles were collected.

[12] Our goal here is not to provide the reader with extensive details on what these models are and how we use them (see the appendix for additional details and references), but rather to provide general intuitions about the use of the LLMs, to understand the building of our corpus.

words they use but by their underlying ideas. For example, the same term—such as "model"—may carry different meanings depending on the surrounding context (*fashion model* vs *scientific model)* and a LLM can detect these variations. By converting words and sentences into vectors that reflect their semantic usage, LLMs make it possible to treat ideas and conceptual shifts as measurable objects, thereby opening new possibilities for the quantitative study of economic thought.

We rely on Sentence-BERT (Reimers and Gurevych 2019), an LLM designed specifically to produce sentence embeddings, that is, numerical vectors that represent the meaning of a sentence. The model assigns one vector to each sentence, which allows for straightforward semantic comparison: sentences that express similar ideas end up with vectors that are mathematically close to each other. Using Sentence-BERT, we vectorized more than 61 million sentences published between 1900 and 2009 from our database. Starting from sentences including "rationality" and "rational", we then searched for the sentences most similar to these anchor sentences. Thus, if sentence A explicitly uses the word "rationality," and its vector is close to that of sentence B—which never mentions the term—sentence B likely discusses a related idea.

Our approach does not simply consist of finding sentences that are close to the words "rationality" and "rational" over a 110-year period, though. Because our project is historical, it is crucial to consider how LLMs themselves handle historical language. LLMs are trained on billions of texts, but the overwhelming majority of these texts are recent.[13] As a result, they tend to offer a "presentist," numerical view of language. For example, current models struggle to reproduce writing styles from earlier periods and cannot reliably infer the publication date of a text (Underwood, Nelson, and Wilkens 2025). Sentence-BERT is subject to the same limitations, and the sentence embeddings it produces inevitably reflect this bias. Besides, as illustrated by Figure 1, our corpus is exponential according to years and most sentences come from recent articles. Taken together, these effects make it very likely that the sentences the model identifies as closest to "rationality" will predominantly come from recent articles. To limit this presentist effect, we adapted the way we compare sentences over time. Instead of comparing a sentence from, say, 1910 directly to *all* sentences in our corpus that contain the words "rationality" or "rational," we constructed a "representative vector" that is a moving average vector with a five-year window. Concretely, for the year 1910, we averaged the vectors of all sentences containing "rationality" or "rational" from 1905 to 1915. This produces a period-specific reference point that reflects how these terms were used *at that particular moment in time.* A sentence from 1910 is therefore judged similar not to the general, and likely modern-day meaning of rationality, but to the way the concept was expressed during its own historical period. This helps ensure that our analysis is sensitive to historical changes in language. Since each document is a set of sentences, we can also identify the documents—and the authors—that discuss rationality most intensively or frequently. Table 1 and Table 2 illustrate respectively the closest sentence and documents (represented by the mean of its vectors) from our representative vectors in 1910.

Exploring the corpus

Simple exploration

Once the corpus has been assembled—which, as we have explained, is far from a "simple" task—the next challenge is to identify how the concept of rationality is used within these texts and how its uses evolve over time.

[13] Moreover, the texts used to train these models are not primarily academic articles in economics. This means that there may be important gaps between the general language patterns the model has learned and the specific vocabulary, concepts, and writing practices of our "domain" (see e.g. Zhang, Rojcek, and Leippold 2025).

Quantitative approaches come in many forms and levels of complexity. Indeed, quantification does not need to be sophisticated to be useful. Simple indicators may not always provide definitive answers to research questions, but they can play an important exploratory role: helping researchers refine their questions, identify anomalies, or detect unexpected patterns.

For textual data, one of the most straightforward indicators is term frequency, that is counting how often particular words or expressions appear. This metric has been used repeatedly in the literature, especially to track the emergence or decline of fields within economics. In well-delimited domains, term frequency can serve as a reliable proxy for intellectual dynamics. For instance, Truc (2023) shows that counting occurrences of highly specific neuroeconomics terms—such as "striatum" or "prefrontal"—closely approximates more advanced quantitative measures, and thus provides a simple but meaningful signal of activity in the field.

For our study or rationality, frequencies of words "rational" and "rationality" already offer important insights. Figure 2 shows the relative frequency (with respect to the total number of words published each year) of "rational" and "rationality." The figure reveals a clear surge in the 1980s and 1990s, suggesting a rise in the mobilization and discussion surrounding relates to the rise of "rational expectations" in macroeconomics and the emergence of behavioral economics with the publication of (Kahneman and Tversky 1979). While we can't rationality. In 1981, the word represented XXXX% of all words, XXXX more times than in 1970. Obviously, we can't easily infer from such simple metrics what it means, but it strongly know what drives this rise, it certainly signals an important moment for the evolution of rationality in economics in terms of intensity, thus prompting for more focused investigation. Another important result from this simple graph is that we only find one large anomalous surge associated with a period where rational choice theory begins to be contested. We could have expected another period of intense controversy during the 1920s followed by a "normalization" of rationality. Instead, we find a slow rise of the occurrence of "rational" words between the 1900s and 1960s signalling a slow and progressive adoption rather than a sudden shift.

To probe this simply dynamic further, we can also investigate how the term is used in specific context with simple metrics. Our aim is to move beyond keyword retrieval and recover the different meanings and intellectual settings in which "rationality" and "rational" are invoked. A direct way to begin is co-occurrence analysis: examining the words that most often appear immediately before or after our keywords. Such co-occurrences indicate the conceptual frames and debates in which the term is embedded.

Figure 3 reports, by decade, the five words most frequently adjacent to "rational" and "rationality," showing how associations shift over time. Before the 1940s, the picture was heterogeneous, with links to philosophy, psychology, and general notions of reasoning (e.g., "rational conduct", "rational organization"). From the 1940s onward, the rise of choice theory places rationality at the center of economic modeling as a device for describing and formalizing behavior. This is mostly visible with the rise of multiple common bi-grams (i.e., combination of two words) that remain stable from the 1930s through the 1980s such as "economic rationality", "rational choice", "rational behavior". By the 1970s—and especially the 1980s—the framework was both extended and contested. First, the most common bi-gram by far becomes "rational expectations" signalling the emergence of a new predominant concept extending rationality.. During the 1980s "rational expectations" appeared XXXX more times than the other most common bi-grams. Second, while we generally associate the rise of "bounded rationality" with Herbert Simon in the 1950s, we can observe that the concept only became prevalent during the 1990s with the bi-gram becoming the third most common one in the 2000s. The rise of "bounded rationality" is also accompanied by a relative decreasing importance of "rational expectations" itself. While it remains the most common bi-gram even in the 2010s, it decreased from XXXX times the second most common occurrence in the 1980s to XXXX in the 2010s.

Beyond textual analysis, another important source of information is documents metadata (e.g., authors, journals, institutions, citations). We focus here on citations to study how articles on rationality cite and are cited. The evolution of an idea often depends as much on how it is circulated and appropriated by readers as on how it was formulated by its authors. A large literature on Citation Theory has developed around how to interpret citations as a scientific practice and its measures (**TahamtanCore1979?**). A wide range of factors influence citation practices. From a genuine desire to acknowledge intellectual debt to more strategic considerations aimed at persuading readers or satisfying peer reviewers, but at the very least citations indicates a relationship that traces between ideas' lineage and their diffusion even if its basis is not only intellectual.

One major structural issue for quantitative analysts using metadata is temporal: systematic citation practices are relatively recent, so citations present poor quality and reliability before the 1960s. In contrast to textual analysis, thus, bibliometric analysis in economics is largely confined to the postwar period. A second constraint is data scarcity and quality: extracting and standardizing references at scale is complex and imperfect. For this reason we rely on Web of Science for structured citation data, despite its proprietary cost and access limits. We then augment our JSTOR full-text corpus with Web of Science citations and with citation and abstract data for key journals missing from JSTOR.

Simply counting citations can be used to proxy engagement and better understand the impact of a particular author or publication. There are clear reasons to incorporate citation studies into historical research. Beyond tracking diffusion, highly cited papers are more visible and attract more engagement (Matthew effect). Recent work also suggests that citations shape reading behavior and perceived quality: highly cited papers are more likely to be read closely and to be seen as substantial intellectual influences (Teplitskiy et al. 2022).

Historians routinely discuss scientific influence, and recognition using proxies such as major grants, prizes, and honors. For example, Sent (2004) narrative of the transition from the dominance of rational choice, through the limited success of "old" behavioral economics (e.g., Simon, George Katona), to the rise of "new" behavioral economics is organized around such milestones. In this sense, citations serve as a complementary proxy alongside traditional markers. For example, while both Simon and Kahneman received the Nobel, they had very different impact on the trajectory of economics in terms of scale, something visible in part by citations. In a large qualitative–quantitative study of the Nobel Prize, Offer and Soderberg (2016) distinguished several profiles: laureates who peak at the prize then decline, "innovators with staying power," "still rising" winners honored before their citation peak, and late winners recognized long after their peak. Relating institutional rewards to citation and publication patterns clarifies how recognition interacts with diffusion, reception, and appropriation.

Figure 4 plots citation patterns for four seminal references on bounded rationality across all economics journals and the top five. The selection is partly arbitrary but standard in the literature: Simon (1955) and Allais (1953) are early critiques of neoclassical rational choice with both empirical and normative implications, while Akerlof (1970) and Kahneman and Tversky (1979) are early "new" behavioral landmarks that mark the emergence and growth of what is simply known as "behavioral economics" for economists.

The influence of "new" behavioral economics overwhelms that of "old" behavioral economics, in both the top five and the full set of economics publications indexed in Web of Science. Whereas Simon (1955) and Allais (1953) are never cited by more than 0.20% of all economics-article, Kahneman and Tversky (1979) reached at least 1% by the late 2010s and continues to rise. The contrast is both in scale of influence and speed of acceptance: citations to the two "new" behavioral papers grow rapidly and steadily right after publication, whereas Simon and Allais peak around the time of their Nobels or only much later in the 2010s wit the emergence of "new" behavioral economics. As argued by Offer and Soderberg (2016), many laureates receive a “Nobel premium,” a modest post-prize citation bump. Simon and Allais fit this pattern, but for Kahneman and Tversky the effect is extreme: after the Nobel, their declining trend reverses and climbs throughout the sample (Offer and Soderberg 2016).

Whether we talk about metadata or textual analysis, such simple tools are perfect because they can be learned quickly by any historians and mobilized in a few days to complement an otherwise qualitative perspective. However, it is also possible to deploy more advanced (and often custom) tools that address specific questions that the researchers has in mind. With this type of tools, the role of quantitative analysis in a given study can shift from complementary to being the leading method of a study (although not necessarily).

More advanced exploration

Citation counts are a blunt tool and can be extended in several ways. One can examine *who* cites a work—by discipline, journal, or subfield—and assess the *qualities* of citations, i.e., distinguishing positive from negative citations or functional roles in the text (litterature review, methodology…) (Budi and Yaniasih 2023). Another way to use citations is to transform citation data into relational data. A prominent example commonly used in the history of economics is bibliographic coupling. Coupling maps relationship between articles by using the similarity of reference lists: articles are nodes, and the more references two article share, the closer they appear in a two-dimensional layout. The premise is that shared references proxies intellectual proximity. These maps help delineate the frontiers of disciplines, fields or sub-specialities and expose their internal organization, including hierarchical and/or core-periphery structures. Using cluster detection algorithm, we identify and groups sets of documents that share a substantial fraction of references and thus have a similar intellectual background. **Our bibliometric clusters thus group documents that are likely to engage with rationality in similar ways based on citations patterns.**

Similarly with textual analysis, word counting and textual co-occurrence analysis offers a first view of how talk about rationality changes, but it is limited to the immediate lexical neighborhood of a word and thus to proximity in vocabulary, not meaning. Two sentences may share no terms—“rational behavior” and “profit maximization”—yet point to closely related conceptions of agency and choice. To move from this semantic approach to a conceptual approach, we turn to recent advances in natural language processing: large language models (LLMs). These models encode context and meaning, allowing us to compare sentences that are semantically similar even when their vocabularies diverge, and to see how the same word takes on different meanings across contexts. **This approach allows for sentence-clustering to identify shared conceptions of rationality across various economics texts and to track large shifts in the way the concept is used over time.**

At the fundamental level, both bibliographic clustering and sentence clustering do a similar operation: **they bring some form of internal order to a large corpus**. Using both methods we are able to identify subgroups within our corpus across multiple dimensions: different sub-fields of economics (e.g., behavioral economics, macroeconomics), different conceptions of rationality (e.g., bounded rationality, rational expectations) or even more heurestically different methods (e.g., experimental, econometrics).

Both methods can be used independently depending on the question at stake, but in our case, they are interdependent and complementary. First, running bibliographic coupling begins with delimiting a relevant corpus: which articles count as being “about rationality”? Keyword searches are brittle, since papers on revealed preference or prospect theory may not use the words “rational” or “rationality.” Using our LLM sentence embeddings from JSTOR we are able to determine to which degree an article engages with rationality in its content independently for the occurrence of the word. For our purpose, we retain the top 10% articles that engaged the most intensively with rationality and formed a corpus suitable for bibliometric coupling (see Appendix XXXX). Second, and as a direct followup to that, both method allows for a variety of quality-cross check. Running bibliometric coupling on a corpus created using our sentence embedding allows us to navigate our copus in a structured manner and make sure that it work as intended. In addition, the systematic comparison of sources and methods is a fundamental principle of historical analysis, needed to triangulate information. Comparing the results from bibliometric and textual clusterizations allows us to identify large trend that are true when looking both at textual patterns and citations patterns. Moreover, because we used two different databases for our two methods (JSTOR for textual analysis and Web of Science for citation analysis), this add another dimensions for triangulations of our results.

Interpreting "Results"

Like with simple tools, advanced quantitative methods do not yield objective, ready-made outputs. In our case, they produce statistical groupings—bibliometric communities and textual clusters—that require qualitative interpretation. This is a classic strategy to reduce a large and complex set of texts to a manageable number of simple categories. But these clusters are constructed on the basis of statistical commonalities, whose actual conceptual meaning can only be characterized through qualitative assessment. More importantly, a good qualitative knowledge of the corpus and of intellectual debates at stake are indispensable to make sense of raw results. **Thus, whether in the collection of data or their analysis, quantitative methods require a constant back-and-forth with qualitative reasoning.** This is especially important because our methods are "unsupervised." With no prior labels to guide classification, human interpretation is central to assessing validity and usefulness.

An example of a results from this analysis stems from "bounded rationality" related clusters. The cluster 40 from the textual analysis is a cluster that exists between 1950 and 2019. We find:

- Wide range of concepts related to bounded rationality: "bounded rationality", "collective rationality", "procedural rationality".
- Herbert Simon as the most recurring author in number of sentence which capture the strong relationship to "old" behavioral economics of the cluster, but also Robert Sugden as a "new" behavioral economists, or Gary Becker as an example of a proponent of rational choice theory who heavily engaged with irrationality
- A mixed of top mainstream journal (The American Economic Review, Econometrica) with more heterodox or specialized journals (Cambridge Journal of Economics, Journal of Economic Issues)
- Example of sentences:
    - Althought has long been agreed that traditional economic theory "assumes" rational behavior, at one time there was considerable disagreement over the meaning of the word "rational." (Becker,

    1962)
    - [R]ationality in real life must involve something simpler than maximization of utility or profit. (Simon, 1959)
    - The various questions that have been raised about the rationality assumption appear to have legitimized and encouraged the development of economic theories

> that model departures from economic rationality in specific contexts. (Kahneman, 2003)

While it grows rapidly in size over time, this textual cluster more generally capture the general controversies in economics surrounding rationality as framed as an “rationality vs irrationality vs bounded rationality” problem in economics. This is well captured by the mixed of authors but also of core dominant journals with more frontiers and heterodox journals.

At the opposite, the bibliographic analysis reveal multiple clusters that capture sub-fields of economics structured by bounded rationality. For example clusters (1) “XXXRISK” and (2) “XXXXSOCIAL” are characterized as follow:

- Wide range of concepts related heuristics, biases and the experimental method: (1) “experimental”, “risk-aversion”, “endowment effect” and (2) “reciprocity”, “fairness”, “ultimatum”, “cooperation”, “trust”.
- The clusters are structured by the work of new behavioral economists like Kahneman, Tversky, Richard Thaler, Colin Camrer, Matthew Rabin or Ernst Fehr.
- Top mainstream journals (The American Economic Review, Econometrica) mixed with specialized journals (Journal Of Risk And Uncertainty, Journal Of Economic Behavior & Organization, Games And Economic Behavior, Experimental Economics)

This time the cluster is not structured by a large debate about bounded rationality, but by the more specific approach of “new” behavioral economists understood as the heuristic and biases research program. Even more precisely, the two clusters make a distinction within “new” behavioral economics between economists working on pro-social behavior and those working on risk and uncertainty. Contrary to textual analysis, we find no real cluster centered around the work of Simon besides a few exceptions like another cluster around operations research. While the textual clusters capture a particular way to discuss rationality in economics which bridge “old” and “new” behavioral economics by this shared interest, the bibliometric cluster analysis reveal the emergence of a sub-fields in the profession with specialized journals and a small set of tightly connected authors that are very different from Simon’s original contribution.

At this point, the clusters only brings some internal order to our corpus and highly different structural patterns in the way rationality has shaped economics whether via its discourse or by the emergence of specific communities. From this type of quantitative results, it is possible to extract historical narrative and develop historical claims by bringing together more traditional approaches in history of economics thought, the historical litterature with our quantitative results.

Discussion

How can these complex quantitative methods and their careful interpretation contribute to our understanding of the history of rationality? How can they help enrich, complete, and refine our understanding of the various and evolving meanings of rationality in economics? This paper does not propose an alternative history of the concept, nor a comprehensive account of its evolution. Its aim is to show, concretely, the benefits, limits, and uses of quantitative methods in the history of economics.

We highlight selected findings that confirm and strengthen strands of the existing literature, broaden its scope, and open paths for further research. We combine multiple indicators with qualitative analysis, reading articles the models pointed out. Our goal is to show how we navigate results and triangulate indicators to produce clear insights and coherent narratives.

Retrieving general patterns from the history of rationality

An obvious way we can use our results is by looking at long term general patterns, something that can be difficult without using quantitative methods. This has two advantages. First, we can write long-spanning histories by focusing on the macro-level shifts in economics as a whole rather than reconstructing it from patchwork of different contributions. Second, even when focusing on sub-fields or specific issues relating to rationality, we are able to recontextualize narrow narratives in a larger picture.

An example of a first big picture shift concerns the evolving relationship between economics and psychology at the beginning of the 19th century. A common narrative is that psychology was "in" during the neoclassical revolution, then "out" with the ordinal and revealed preference revolution," then "back in" with the behavioral and experimental economics (Giocoli 2003; **handsEconomics2010?**). These large movement can be well captured by synthetic metrics like citations between disciplines (Truc 2023) but it can be quite difficult to connect to the historical literature like Giocoli (2003) who analyze big figures in economics such as Vilfredo Pareto, Irving Fisher, Lionel Robbins, and Paul Samuelson, to reconstructs the long-standing debate over the relation between economics and psychology. With our textual analysis we are able to identify these different episodes by large shift in how rationality is used and discussed, but also by whom they are discussed.

The movement of psychology "in" economics is characterized by the prevalence of hedonistic psychology in economics. This debate about the role of hedonism in explaining behavior appears clearly in Cluster 2 (**?@fig-llm-alluvial**), which extends from 1900 to the 1940s. This cluster is characterized by terms that completely disappeared from contemporary economic vocabulary: "satisfaction", "pleasure", "instinct", "human nature", "pain". The cluster brings together advocates of hedonistic foundations and their critics, notably institutionalists who questioned psychological grounding. Wesley C. Mitchell (1916, 160) illustrates this opposition in one of the closest sentences to our representative vector: "to find the basis of economic rationality in the development of a social institution directs our attention away from that dark subjective realm, where so many economists have groped, to an objective realm, where behavior can be studied in the light of the common day".

As the hedonistic cluster disappear in the 1940s, it is replaced by multiple clusters that capture the shifting away from psychology toward perfect rationality in economics. This is captured by cluster 31 in the 1940s, notably Friedman and Savage (Friedman and Savage 1948), and especially after 1950 in cluster 46. This clusters reunite the debates about demand theory, the use of cardinal utility, and the possibility of interpersonal comparisons. Chicago economists such as Friedman and Savage (Friedman and Savage 1952), Stigler (Stigler 1950), and later Becker (Becker 1962) are prominent. In parallel, Nicholas Georgescu-Roegen offered a sharp critique of "utility" and its measurability (Georgescu-Roegen 1954). The notion of perfect rationality also structured many other related clusters such as the ones about price theory (cluster 28, 1940 to 1989) or profit maximization (cluster 30, 1940 to 1969) where rationality becomes a more standardized concept embedded in particular economic theories.

From the 1980s onward, challenges to the standard rationality approach embodied by expected utility theory emerged in several textual clusters (e.g., clusters 83 and 93) and persisted until the end of our period. A distinct behavioral economics cluster (106) appeared in the 2000s. This trend is confirmed by the bibliometric analysis that shows a first community labeled "Behavioral Economics and Choice Theory" in 1977–1984, followed by many communities after 2000 that form a dense network around the issue of how to model rationality (**?@fig-biblio-alluvial**), something we will explore further in the following sections.[14]

[14] See for instance the communities "Behavioral Economics and Experiments", "Ambiguity and Uncertainty", "Intertemporal Choice and Control" or "State Preference Valuation".

Another big picture shift is the relationship between macroeconomics and microeconomics. While the history of the relationship between economics and psychology even as told by Giocoli (2003), (**handsEconomics2010?**), or Heukelom (2014) mostly focus on microeconomics, our approach embedded this individual rationality history into a larger scope. For example, a structural distinction in our quantitative analysis is between individual-focused clusters and more macro or general economics clusters.

For example, as psychology is driven out economics and the discussions surrounding rationality become less about hedonistic concepts (pleasure, instinct), and more about economic concepts (profit, consumption…), we find more economics-focused clusters. The 1940s and 1950s reveal two textual clusters centered on debates over economic freedom and free enterprise versus economic planning. These clusters include roundtables and special issues on planning, such as a discussion sparked by Oskar Lange's 1949 article (see Perroux et al. 1949), with contributions by François Perroux, Jan Tinbergen, Evsey Domar, and Michał Kalecki. They also include debates on "the proper spheres of individual freedom and collective control in the 'good' economy" (Taylor 1948), in a collection that included, among others, Henry Simons. Across both clusters, Frank Knight appears as a recurrent reference.[15] Somewhat linked to this is the textual cluster 53, which from the 1950s on discussed social choice and welfare, with rationality framed as a guide for decision makers.[16]

For the most recent period, we find the emergence then domination of "rational expectations" in the way rationality is discussed in economics. In both bibliometric and textual analyses, from the 1970s rational expectations became central and occupied substantial discussion, appearing across multiple textual clusters and bibliometric communities (**?@fig-llm-alluvial**; **?@fig-biblio-alluvial**). More generally, during the 1970s, most clusters are about macroeconomics issues or economics-oriented issues. In the 1970-1977 coupling network, the five biggest clusters are about (1) rational expectations and macroeconomics, (2) finance, (3) public goods and externalities, (4) public goods ; (5) disequilibrium and Keynesian economics. While these clusters use rationality in different ways and with different approaches, we find no clusters dedicated to rationality issue at the individual level. This situation change slowly during the 1980s. In the 1980-1987 network, we find the "Game Theory and Information" (12% of the network) about subgame perfect equilibrium and incomplete information (Reinhard Selten, George Akerlof, John Harsanyi), and the "Behavioral Decision Theory" clusters (2% of the network) capturing the beginning of behavioral economics as promoted by Kahneman and Tversky.

Taken together, these individual-rationality clusters that did not exists in the 1970s now represent 12% of the network. This share rose to around 29% of the network in the 1990s and around 40% of the networks in the 2000s. While existing history of rationality often focus on the microeconomics side or the macroeconomics side, we are able here to measure a large shift in where rationality is discussed. During the 1970s, as psychological approaches were marginalized, rationality remained primarily embedded in sub-fields concerned with markets and macroeconomic issues. The rise of behavioral economics and new approaches in game theory fundamentally changed this dynamic, leading to a specialized investigations of individual rationality that now dominate. This transformation represents not just a change in research topics, but a fundamental reframing of how economics approaches the concept of rationality itself from an assumption embedded in aggregate models to an object of direct investigation at the individual level.

These general patterns matches mostly known patterns in the history of economics but with the advantage of situating them in a larger context and weight their respective importance and relationship to each others. However, it is also possible to highlight a more surprising results that is less discussed in the historical litterature.

[15] See also textual cluster 31 on planning issues.

[16] Kenneth Arrow, Herbert Simon or James Buchanan are central in these debates.

Beyond such long-spanning histories, it is also to focus on specific issues to confront more explicitly existing historical narrative and cross-check our quantitative results with more qualitative investigations. In the following sections we focus on two issues in the historical literature with the "rational expectations revolution" and the "birth of behavioral economics".

A rational expectations revolution ?

One focused read of our results centers on "rational expectations." The concept has an early history: developed by Muth (1961) for price movements in agriculture; popularized when Carnegie colleagues—Robert E. Lucas, Edward C. Prescott, and Thomas J. Sargent—applied it to macroeconomics (Robert E. Lucas Jr. 1972; Thomas J. Sargent and Wallace 1973). It became central in the early 1970s and quickly controversial for monetary and fiscal policy (see e.g., Thomas J. Sargent and Wallace 1975). By the late 1970s it circulated in policy and the press and was often described as a "rational expectations revolution" (Duarte 2025).

Our methods help account for the rapid diffusion beyond controversial policy debates during the stagflation era. First, the results show no significant trace of rational expectations in the 1960s in either bibliometric or textual outputs.[17] From the 1970s it has taken a central place in the literature on rationality.

The controversial character of rational expectations appears first in the text analysis. Two clusters on rational expectations emerged in the 1970s (textual clusters 67 and 72). The first gathers articles that debate the hypothesis itself, including many critiques of its theoretical and empirical relevance in the 1970s–1980s. Others examine the implications of adopting the hypothesis across domains. The second focuses on models that use rational expectations rather than the hypothesis per se.[18] Here, alongside promotion of such models—Sargent is a major contributor—there are early critiques targeting both the rational-expectations assumption and additional auxiliary assumptions. A clear example is Ray Fair's complaint about "one class of macroeconomic models that have recently been developed" which rely on "(1) the assumption that expectations are rational, given the available information; (2) the assumption that information is imperfect regarding the current state of the economy; and (3) the postulation of an aggregate supply equation in which aggregate supply is a function of exogenous terms plus the difference between the actual and expected price level" (Fair 1978, 411).[19] Diffusion is also visible in clusters that predate the use of rational expectations but incorporate it from the 1980s onward, notably cluster 76 on monetary economics and cluster 28 on price theory.[20]

Our bibliometric analysis complements the text analysis. By tracking references and using overlapping windows, it detects the emergence of new communities and the split of others with fine temporal resolution. The main rational-expectations community, "Rational Expectations and Business Cycles", first appeared in 1966–1973. It focuses less on the hypothesis itself and more on its macroeconomic policy implications. The controversy centers on modelling the inflation–unemployment trade-off and the implied (in-)effectiveness of

[17] This highlights a key caveat. Because these methods foreground statistically "significant" patterns and indicators' prevalence, they are not always well suited to tracing and interpreting the origins of a concept or theory with the care that close historical study and archival research provide.

[18] The second textual cluster is less centered on rational expectations than the first. It aggregates model-focused debates—some on rational behavior in general, others on demand—within which rational-expectations models remain prominent.

[19] The mix of defense and opposition continues into the 2000s–2010s: we see discussions of DSGE models , alongside macro models with bounded rationality or learning and arguments for behavioral macroeconomics and rational inattention.

[20] Other clusters linked to rational expectations cover inflation expectations and the term structure in the 1970s (cluster 78), forecasting (81), wages and unemployment (82), finance (87), information (89), econometric issues (92), and a more theoretical intersection of rational expectations and game theory from the 1980s (88).

monetary policy. This aligns with the early divide between "old" and (future) "new" Keynesians (Goutsmedt et al. 2019).

The controversial status of rational expectations in the 1970s–1980s likely helps explain the prominence of this community in our results. These controversies are well documented (Hoover 1988; Goutsmedt et al. 2019). At the same time, our evidence suggests diffusion by application across domains. The bibliometric analysis indicates two broad pathways. First, communities initially outside the "Rational Expectations and Business Cycles" community began to incorporate the assumption. Second, new communities gradually branched off from the core community: articles that first co-located with the main group increasingly coalesced into autonomous clusters.

Focusing on the first pathway, one community engaged with rational expectations slightly earlier than the "Rational Expectations and Business Cycles". This community addressed topics in trade, demand, and investment within a general-equilibrium framework. Muth (1961) belongs to this community and remains a key reference, even though the rational-expectations hypothesis is not central. In 1971–1978, the label shifts to "Investment and Economic Growth," where Lucas's early work on investment is influential, albeit without using rational expectations. Several important references in this window do employ the assumption, including Cyert and DeGroot (1974) and Townsend (1978). In 1975–1982, the community transformed into a new community, "Investment and Uncertainty," now featuring Kydland and Prescott (1977), Kydland and Prescott (1982), and Robert E. Lucas (1978).[21] Thus, beyond the well-known debates on business cycles and inflation, parallel communities—only partly focused on macroeconomic issues—were active in the early 1970s.

After the mid-1970s, several independent communities began to adopt rational expectations. A first trajectory appears in 1976–1983: two finance-oriented communities ("Asset Pricing and Consumption" and "cl_319") partially merged into "Rational Expectations and Market Information." At its core was imperfect information, with Rothschild and Stiglitz (1976) as a key node. Another central contribution was Grossman and Stiglitz (1980), which drew on Lucas's imperfect-information framework (Robert E. Lucas Jr. 1972) and combined rational expectations with noisy signals to question market efficiency (see Delcey and Sergi 2023). A second community emerged in 1979–1986 with "Game Theory and Information," where rational expectations became more prominent. Here Kydland and Prescott (1977) and Barro and Gordon (1983) on time inconsistency sit alongside Kreps and Wilson (1982) on sequential equilibria and Selten (1975) on equilibrium refinements. In the early 1980s this community split, yielding "Monetary Policy and Inflation," which leverages game-theoretic models to deal with credibility and reputation in policy design.

A second diffusion process involves the emergence of new communities organized around rational expectations that gradually separate from the initial core. In the early 1970s, an international macroeconomics community branched off from "Rational Expectations and Business Cycles." Although it addressed a range of international macroeconomic topics, the determination of exchange rate dynamics—and the role of the rational-expectations hypothesis in that determination—quickly became central, notably with Dornbusch's overshooting model (Dornbusch 1976). From 1974–1981, another community grew out of the core—"Inflation, Expectations and Interest Rates"—which concentrated on the term structure and the use of interest rates to forecast inflation. This community became a meeting ground for macroeconomics and finance, where rational expectations and efficient markets were closely linked (Delcey and Sergi 2023).

This issue of the diffusion of rational expectations merits a dedicated study. Here, the goal is to illustrate the diversity of trajectories and the fine-grained mapping our analyses produce, enabling productive interaction between quantitative evidence and qualitative interpretation.

[21] These articles received a good number of citations from the paper of this community, but they appear as "connector" in the sense that they were highly connected to nodes in other communities.

By the early 1980s, roughly half of the network consisted of communities engaging, to varying degrees, with rational expectations. At the same time, a behavioral economics community had emerged, though it still formed a small share of the network, a situation that changed after the 1990s.

Simon's reception and the birth of behavioural economics

A second focus reading of our study concerns the birth of behavioral economics. The term "behavioral economics" has a complex history, with George Katona often credited as an early adopter (**giladEconomic1984?**; **hosseiniGeorge2011?**). Sent (2004) made an historical distinction between "old" and "new" behavioral economics to separates the "old" heterogenous attempts by Katona, Simon and others to reform mainstream economics from the "new" more homogenous heuristic and biases research program of Kahneman and Tversky. Sent argues that "new" behavioral economics succeeded precisely because it worked within the existing paradigm rather than challenging it fundamentally. Additionally, it emerged in the 1980s when economic theory faced multiple epistemological challenges, creating an opportune moment for alternative approaches. While our method cannot pinpoint the exact epistemological reasons for this shift, we can trace the reception and influence of both approaches to understand how and when they diverged.

Simon's concept of "bounded rationality" stands as one of the most influential critiques of standard rationality in economics in the sense that it framed a lot of the debates surrounding rationality. His presence dominated 1950s discussions, with his seminal articles and Nobel lecture (Simon 1979) appearing across numerous research areas from social choice theory to the theory of the firm and price setting.

Our textual analysis reveals Simon's conceptual centrality. Textual cluster 40, spanning roughly seventy years from 1950 onward, captures general discussions on rationality in economics. While early debates (1950s) focused on game theory and expected-utility theory (Schelling 1959; Ellsberg 1956; Marschak 1950; Chernoff 1954), by the 1970s "bounded rationality" became the cluster's most prevalent expression. By the 1990s, discussions of "boundedly rational agents," "unbounded rationality," and "procedural rationality" clearly positioned Simon's concepts as central to economic thought signaling a late but significant integration.[22] This is corroborated by citation analysis as citations to Simon (1955) only rised very slowly in economics after its publications with peak citations happening in the post 2000s (Figure 4).

However, this conceptual influence did not translate into stable research communities. Bibliometrically, we find only small, unstable clusters formed around Simon's work. From 1960–1967, his ideas appeared in operations research circles (cl_5) and organizational theory critiques of profit maximization. Simon his more generally associated with other researcher critical perfect rationality like Winter (1971), which applies satisficing to firm behavior, and Cyert and George (1969). A related strand is Leibenstein (1966) on "X-efficiency," which challenges economics' focus on allocative efficiency. Together, Simon, Winter, Cyert (with George), and Leibenstein form a diverse critique of how rationality—especially through profit maximization—is employed in economics, from an organizational theory perspective. After 1969–1976, this stream landed in a smaller, short-lived community (cl_175). References to Simon and critiques of profit maximization then remain scattered, moving through several small communities (cl_182, cl_251, cl_300, cl_328). While Simon's influence in economics was enduring, it remains marginal.

In the 1977–1984 period, Simon's work was absorbed into a large "Behavioral Economics and Choice Theory" cluster alongside Kahneman's prospect theory. A similar phenomena can be

[22] Simon himself became one of the most prevalent words of the cluster.

observed in the textual analysis as the textual cluster 40 commented earlier. The cluster becomes increasingly populated by "new" behavioral economists (Matthew Rabin, Daniel Kahneman or Robert Sugden) and now includes critical comparison of "new" behavioral economics with other approaches such as the one formulated by Nathan Berg and Berg Gigerenzer (**bergAs-if2010?**) who criticize the tameness of "new" behavioral economcis models.

The crystallization of a distinct "Behavioral Decision Theory" around Kahneman and Tversky's contributions signal the beginning of an explosive growth of "new" behavioral economics. The cluster spins off several autonomous communities capturing the emergence of sub-fields within behavioral economics: pro-social behavior (cl_x), intertemporal decision-making (cl_x), behavioral game theory (cl_x), and behavioral finance (cl_x). By the 2000s, behavioral economics had become the dominant venue for research on rationality, with five behavioral economics clusters representing approximately 40% of the entire network.

The contrast with Simon is striking. While "bounded" and "procedural" rationality remain regularly invoked concepts, no later research community clearly carries Simon's heritage as a bibliometric anchor. The scale and speed of acceptance differed dramatically between the two approaches. By 1980, Kahneman and Tversky (1979) was already more cited than Simon (1955) had been at that time (Figure 4). By 1985, prospect theory had surpassed the lifetime citation peak that Simon's work ever achieved in economics. (Figure 4). By the 2000s, most microeconomics publications about rationality are related to behavioral economics, and more generally, most publications about rationality are connected to the research program.

A first surprising results from our quantitative study is the stark contrasting reception of Kahneman and Simon in terms of temporality. For historian Floris Heukelom (**heukelomSense2012?**; Heukelom 2014), a pivotal moment in the history of behavioral economics is the explicit creation of research program within the Sloan and Russell Sage Foundations between 1984 and 1992 that most notably led to the pairing of Kahneman and Thaler that changed the trajectory of behavioral economics. However, our quantitative analysis suggests that even in the early 1980s it was clear that something particular was happening in terms of reception with the work of Kahneman and Tversky. Despite Kahneman and Tversky (1979) being the only economics publication of duo until 1985 and long before the Sloan-Sage Foundation research program (1984–1992) that formalized behavioral economics, prospect theory became very cited rapidly (Figure 4). As early as the 1977–1984 bibliometirc network, we find a cluster structured by the work of Kahneman and Tversky suggesting that not only their article his cited, but it rapidly structured a new strand of research in a way that never happened with Simon's work in economics.

This is particularly remarkable given the external similarities between the two cases. Both Kahneman and Simon published in top economic journals and received Nobel Prizes. While Simon's Nobel only increase his influence in economics marginally, Kahneman's Nobel reversed a downward citation trend in the 2000s to an upward trend cimenting his contribution as a structural pillar of a larger wide spanning research program.

A second surprising result is that citations to Simon (1955) have never been higher than since the ascent of the new behavioral economics. Earl (2022) is critical of how much "old" behavioral economics seems to have been forgotten by "new" behavioral economists:

- "Neither Kahneman nor Thaler have sought to promote earlier behavioral economics alongside more recent work. Instead, they give the impression that behavioral economics started around 1979–1980 with the publication of Kahneman and Tversky's (1979) article on prospect theory and that theory's use by Thaler (1980). All in all, this is a very curious state of affairs: a cynic might suggest that it looks rather as if the earlier work has been airbrushed from the history of economic thought by the strategic redefinition of what constitutes behavioral economics. A more charitable and reflexive

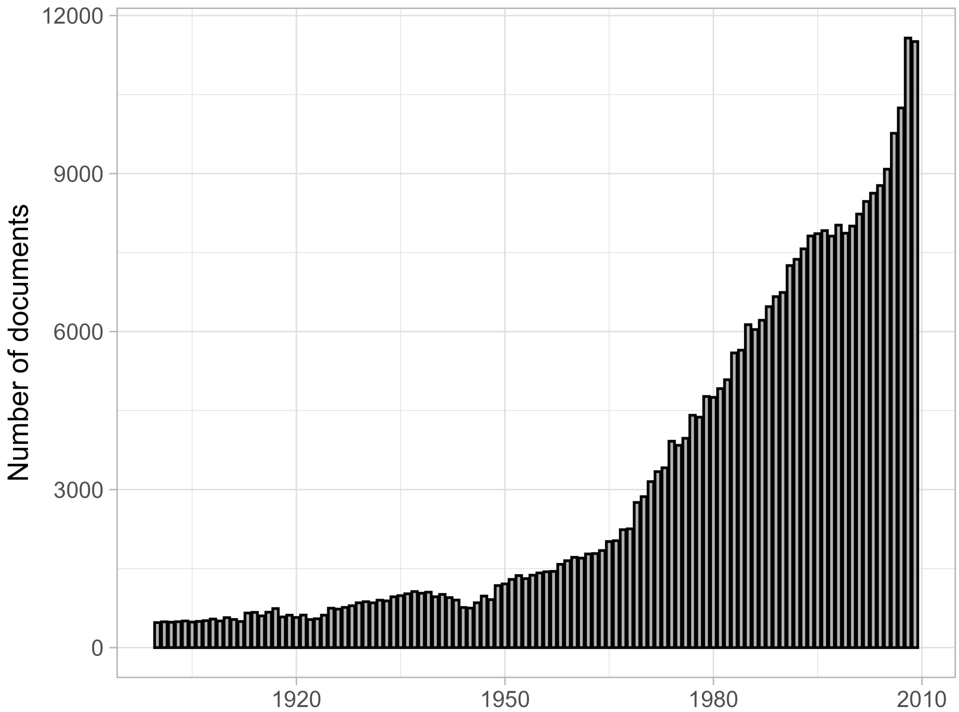
> view would see the situation as resulting from insufficient familiarity with the earlier literature [...]" (Earl (2022), p.2)

The rising citations to older work from Simon or Allais in our example show that while it is possible that most "new" behavioral economics rarely acknowledge the contributions of "old" behavioral economists, the scale of the rise of "new" behavioral economics still led to more attention paid to "old" behavioral economists than ever before in economics. This resurgence admits at least three interpretations. A more favorable reading is that the "new" program helped revive abandoned research directions and moved toward reconciliation with earlier strands (something promoted by Sent (2008) or more recently by Earl (2022)). A more unfavorable view sees intellectual appropriation, in which classic references are reframed to fit the new agenda, renewing interest but with a biased and presentist lens (Mongin 2019). A third more critical interpretation is that the citation increase reflects a growing backlash against the "new" program from the standpoint of "old" behavioral economics. Preliminary evidence supports the second interpretation. This is suggested by the sparse and unstructured citations patterns to "old" behavioral economics and the fact that discussions of rationality are increasingly framed by "new" behavioral and experimental concepts. While a definitive conclusion is beyond the scope of this paper, our study provides a roadmap for future quantitative and qualitative research needed to further the understanding of the relationship between "old" and "new" behavioral economics.

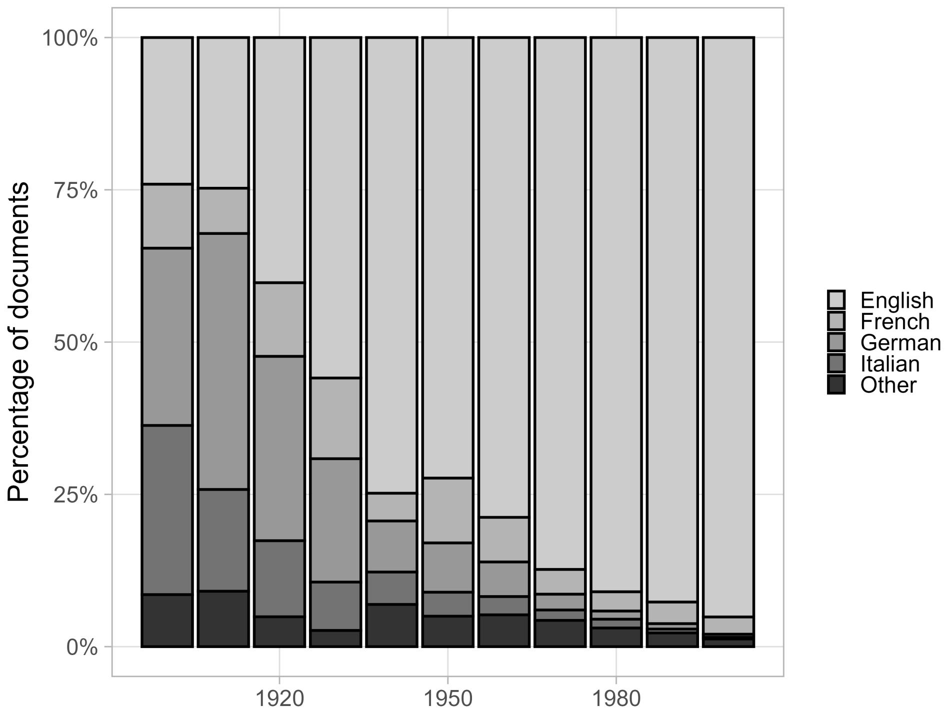
Our quantitative studies delineates the specific chronology and magnitude of the different attempts at stirring a behavioral shift in economics. However, as made clear by Figure 4, it is not just a linear story of a shift from one program to the other. citations to Simon (1955) have never been higher than since the ascent of the new behavioral economics. This resurgence admits at least three interpretations. A more favorable reading is that the "new" program helped revive abandoned research directions and moved toward reconciliation with earlier strands (something promoted by Sent (2008) or more recently by Earl (2022)). A more unfavorable view sees intellectual appropriation, in which classic references are reframed to fit the new agenda, renewing interest but with a biased and presentist lens (Mongin 2019). A third more critical interpretation is that the citation increase reflects a growing backlash against the "new" program from the standpoint of "old" behavioral economics. Preliminary evidence supports the second interpretation. This is suggested by the sparse and unstructured citations patterns to "old" behavioral economics and the fact that discussions of rationality are increasingly framed by "new" behavioral and experimental concepts. While a definitive conclusion is beyond the scope of this paper, our study provides a roadmap for future quantitative and qualitative research needed to further the understanding of the relationship between "old" and "new" behavioral economics.

Conclusion

Figures



(a) Documents



(b) Languages

Figure 1.: Distribution of documents and languages in the full-text database over time (1900-2009).

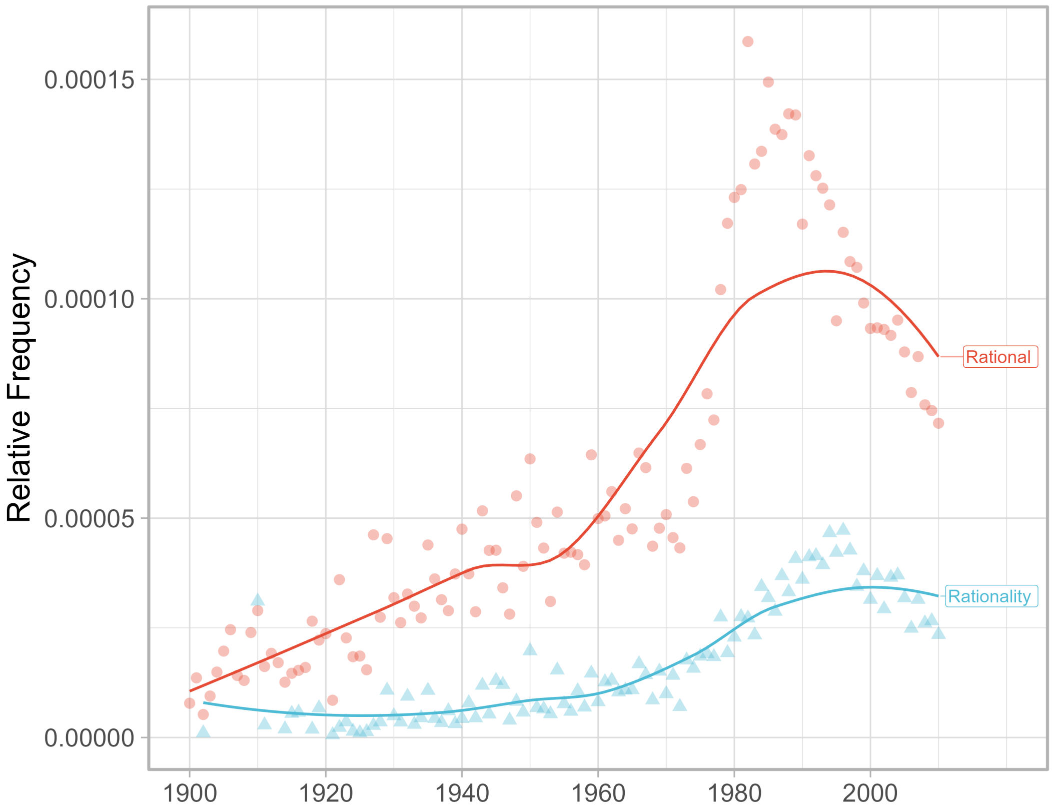


Figure 2.: Relative frequency of terms "rational" and "rationality" in the jstor database (1900-2009). Points are the empirical data. Lines are the loess smoothed trends.

Table 1.: Sentences from the corpus closest to the representative vector in 1910.

Sentence	Cosine similarity
From another side, the concepts supply the basis for rationality.	0.690
The recital is important only as tending to show that a theory of valuation which places the emphasis upon rationalistic appraisal overlooks the most important features of the process which it seeks to explain.	0.686
Complete reliance may be placed upon the rationality of both the pecuniary and the hedonic subject.	0.678
It is exceedingly doubtful whether such hyper-rational view of human nature could have gained wide credence among men not themselves disciplined in the use of pecuniary concepts.	0.670
More fundamental is the great problem of accounting for economic rationality itself.	0.656

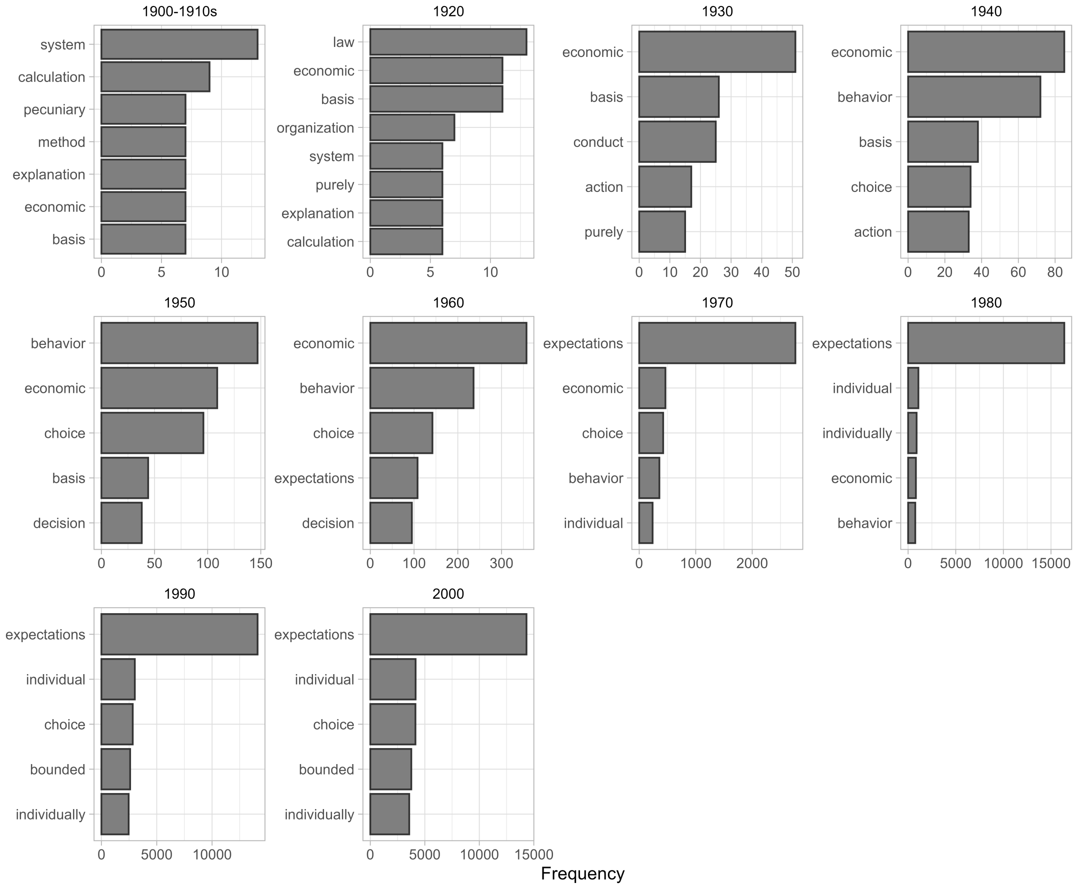


Figure 3.: Most frequent neighbor terms of "rational" and "rationality" by decade

Table 2.: Documents from the corpus closest to the representative vector in 1910.

Title	Authors	Journal	Cosine similarity
The Rationality of Economic Activity	Wesley C. Mitchell	Journal of Political Economy	0.862
Social Productivity Versus Private Acquisition	H. J. Davenport	The Quarterly Journal of Economics	0.824
The Futility of Marginal Utility	E. H. Downey	Journal of Political Economy	0.786
Observation in Economics. Annual Address of the President	Davis R. Dewey	American Economic Association Quarterly	0.778
The Economics of Henry George's "Progress and Poverty"	Edgar H. Johnson	Journal of Political Economy	0.765

Note:
Documents vectors are computed as the average of their sentences vectors.

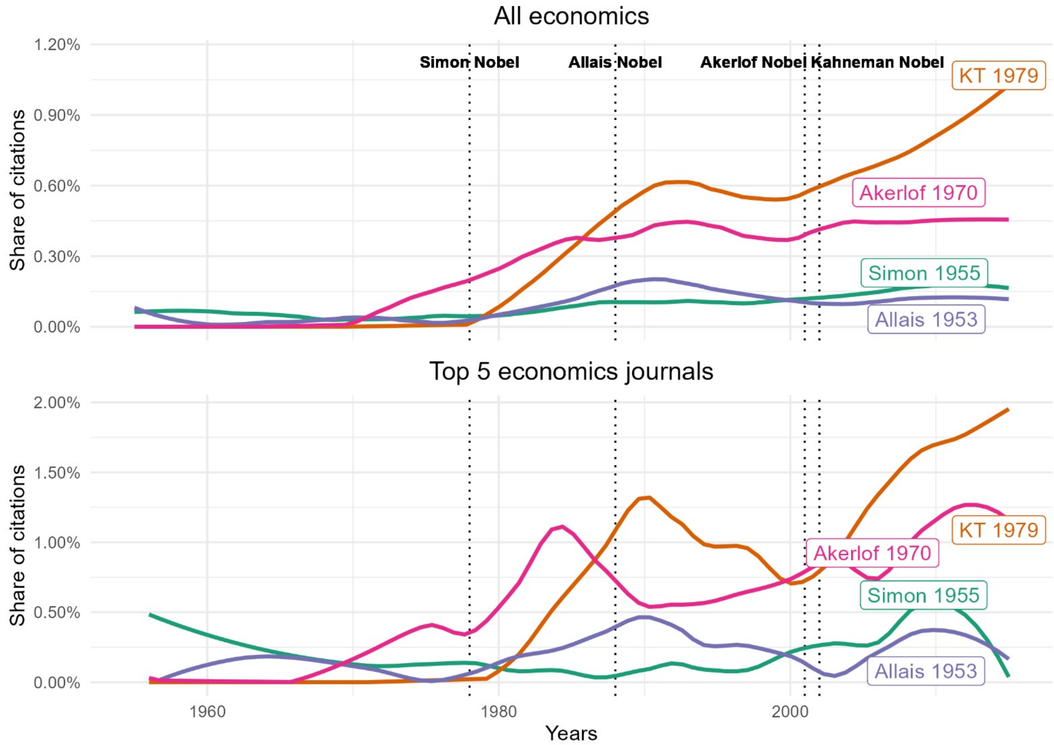


Figure 4.: Relative citation of a selection of seminal papers on behavioral economics

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Appendix

Overview of the data and methods

Here, a quick summary of all the databases used and the methods used. Could be nice to have a visual representation.

Building the full-text database

Making clear the difference between the bibliometric database and the full-text database, and the full-text database and the rationality corpus.

Elaborating the list of economics journals

Extraction and cleaning process for economics full texts

Overview of the full-text database content

Merging with the bibliometric database

Creating a rationality corpus

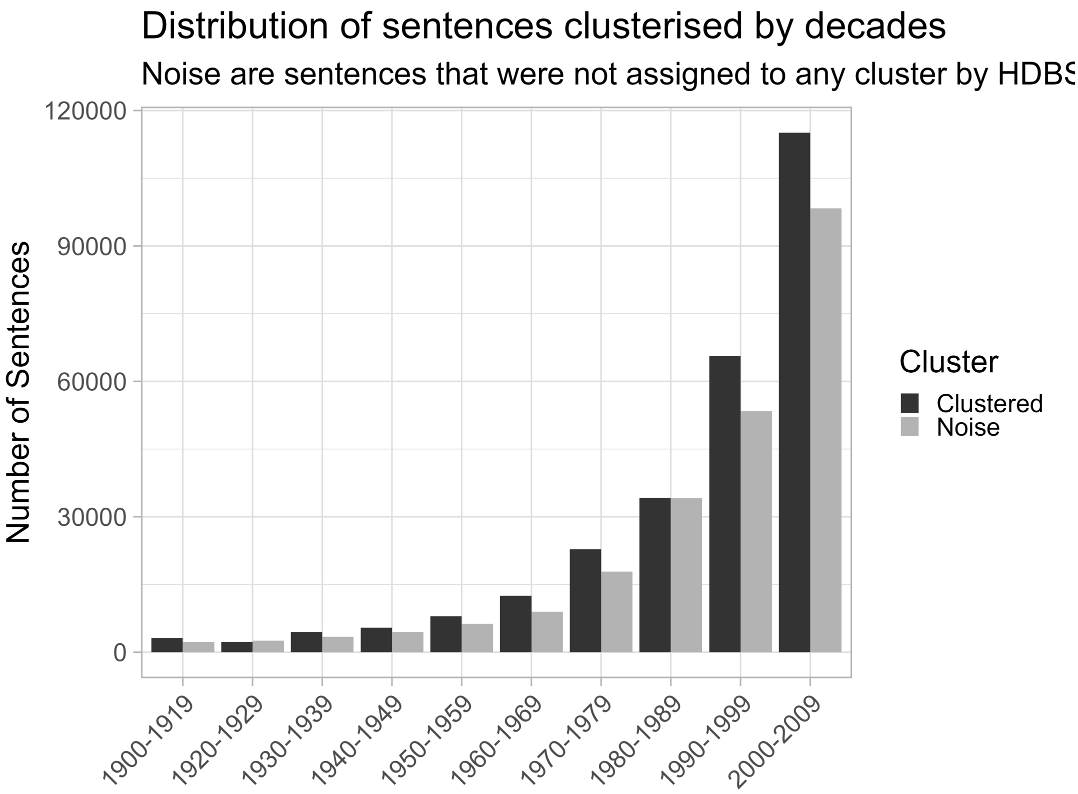
Using LLMs and sentence embeddings to represent sentences

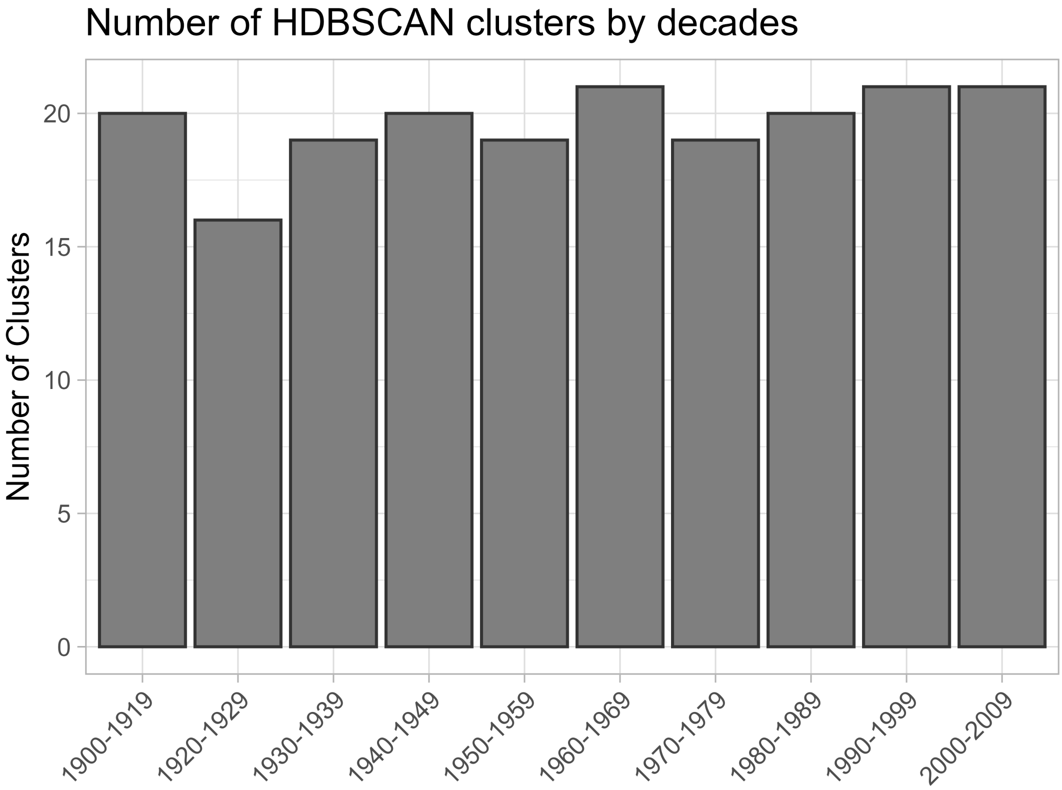
Building the representative vector of rationality

Selecting full-texts and sentences related to rationality

Text clustering methods

Sliding time windows and HDSBSCAN





Backbone networks and matching clusters over time

Bibliographic coupling